

E-commerce Sales Analysis Project

★ **Project Objective:**

The **Ecommerce Sales Analysis Project** focuses on cleaning, transforming, and analyzing ecommerce sales data using Microsoft Excel. The objective is to identify sales trends, customer purchasing patterns, product performance, and profitability insights to support business decision-making.

★ **Dataset Description:**

The **Ecommerce Sales Dataset** contains transactional sales data from an online retail business. It includes information about orders, customers, products, sales performance, discounts, and shipping details. The dataset is designed for practicing data cleaning, transformation, exploratory data analysis (EDA), and business intelligence reporting using Excel.

★ **Column Description:**

Column Name	Description
Order ID	Unique identifier for each order.
Order	Date when the order was placed.
Customer ID	Unique identifier for each customer.
Customer Name	Name of the customer.
Product Name	Name of the product purchased.
Category	Product category.
Quantity	Number of units sold.
Unit Price	Price per unit of the product.
Discount	Discount applied to the order.
Sales/Revenue	Total revenue generated from the sale.
Region/Location	Geographic area of the customer or sale.
Shipping Information	Details related to product delivery.

★ **Data cleaning & transformation:**

Customer Details - Sheet

Data Cleaning & Transformation

1. Cleaned all columns using TRIM(), CLEAN(), and PROPER() functions to remove extra spaces, non-printable characters, and inconsistent text formatting.

2. Formatted column text and numeric values according to the data requirements.

3. To ensure consistency across the dataset, used Find and Replace to standardize values:

- * Transformed Customer IDs from the format C-1486 to CUST1486.
- * Removed honorific titles such as Dr., Mr., and Mrs. from the Customer Name column.
- * Standardized the Country column by replacing United States of America with USA.

Formulas & Functions

Missing values in the Loyalty Level column were identified and replaced with "Unknown" using the IF() and ISBLANK() functions to maintain data consistency.

Formula =IF(ISBLANK(H2),"Unknown",H2)

Outcome

The Customer_Dim dataset was successfully cleaned and transformed using Excel data-cleaning techniques. Standardizing Customer IDs, customer names, country values, and loyalty levels significantly improved data quality. The cleaned dataset is now suitable for reporting, dashboard creation, customer segmentation, and further business analysis.

* Features Used

TRIM()
CLEAN()
PROPER()
IF()
ISBLANK()
Find & Replace
Sort & Filter
Excel Tables
Data Validation

Product Details - Sheet

Data Cleaning & Transformation

1. Cleaned all columns using TRIM(), CLEAN(), and PROPER() functions to remove extra spaces, non-printable characters, and inconsistent text formatting.

2. Formatted column text and numeric values according to the data requirements.

"Number Stored as Text":

Click the warning icon.

Select Convert to Number.

Or:

Select the entire column.

Go to Data → Text to Columns.

Click Finish.

This converts text numbers to real numbers.

3. then we can add the currency symbol in cost column.

4. To ensure consistency across the dataset, used Find and Replace to standardize values:

* Transformed Customer IDs from the format p-3 to PROD3 etc,...

5. Duplicate vales are removed

Formulas & Functions

1. Missing values in the Cost column were identified and replaced finding the AVERAGE(), IF() and ISBLANK() functions to maintain data consistency.

Formula =IF(ISBLANK(F2),AVERAGE(\$F\$2:\$F\$101),F2)

2. Missing values in the Stock column were identified and replaced with the average stock quantity to ensure data consistency and minimize the impact of missing data on analysis.

Formula =IF(ISBLANK(G2),AVERAGE(\$G\$2:\$G\$101),G2)

3. Used VLOOKUP to retrieve the Product Name, Category, Cost, and Stock Quantity based on the Product ID.

Formula =VLOOKUP(I4,A2:G101,{2,3,6,7},FALSE)

Outcome

1. Missing values in the Cost and Stock columns were successfully handled using average-value imputation.

2. Product information was automatically retrieved using Product ID, reducing manual effort and improving accuracy.

3. The dataset is now cleaner, more complete, and ready for further analysis and reporting.

VLOOKUP

Product_ID	Product_Name	Category	Cost	Stock
PROD1	Men's Graphic T-Shirt	Clothing	342.6	181
PROD27	Decorative Floor Vase	Home	254.0	378
PROD35	Bedroom Table Lamp	Home	382.2	329
PROD46	Men's Winter Jacket	Clothing	167.3	296
PROD58	Womens Cotton Scarf	Clothing	113.8	485
PROD65	Adjustable Dumbbells 20kg	Sports	240.6	423
PROD75	Women's Ankle Boots	Clothing	158.1	82
PROD89	Face Foundation Kit	Beauty	183.0	457

Store Details - Sheet

Data Cleaning & Transformation	1. Cleaned all columns using TRIM(), CLEAN(), and PROPER() functions to remove extra spaces, non-printable characters, and inconsistent text formatting. 2. Formatted column text and numeric values according to the data requirements.
Formulas & Functions	The XLOOKUP function was used to retrieve store details based on the Store ID. The lookup returned the Store Name, Region, City, and Store Type from the Store Dimension table, enabling quick and accurate access to store information. Formula =XLOOKUP(H5,STORE_DIM[Store_ID],STORE_DIM[[Store_Name]:[Store_Type]],0)
Outcome	1. Store details were automatically retrieved using the Store ID. 2. Reduced manual data entry and lookup errors. 3. Improved data accuracy and efficiency in reporting.

XLOOKUP

Store_ID	Store Name	Region	City	StoreType
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE13	Gardner, Humphrey and Contreras	East	Johnsonville	Flagship
STORE4	Alexander Group	South	Port Sharon	Online
STORE15	Shelton PLC	North	East Jenniferstad	Flagship
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
STORE9	Moore, Padilla and Butler	South	East Johnstad	Flagship

Sales Details - Sheet

Data Cleaning & Transformation

1. Cleaned all columns using **TRIM()**, **CLEAN()**, and **PROPER()** functions to remove extra spaces, non-printable characters, and inconsistent text formatting.
2. Formatted column text and numeric values according to the data requirements.
"Number Stored as Text":
 Click the warning icon.
 Select Convert to Number.
 Or:
 Select the entire column.
 Go to Data → Text to Columns.
 Click Finish.
 This converts text numbers to real numbers.
3. Then we can add the currency symbol in **Unit Price** and **Total Amount** column.
4. Validation done for **Amount** and **Quantity** column.
5. Used Conditional formatting to find highest sales value.

Formulas & Functions	1. Missing values in the QUANTITY column were identified and replaced finding ith 1 , IF() and ISBLANK() functions to maintain data consistency.
	Formula =IF(ISBLANK(F2),1,F2) 2. Missing values in the DISCOUNT column were identified and replaced finding ith 0 , IF() and ISBLANK() functions and change the column to discount value. Formula =IF(ISBLANK(H2),0,H2) 3. Missing values in the Totla Amount column were identified and replaced finding the AVERAGE() , IF() and ISBLANK() functions to maintain data consistency. Formula =IF(ISBLANK(J2),AVERAGE(\$J\$2:\$J\$2001),J2) 4. The data extracted in July 2024 , the total revenue reached \$2,175,265 as calculated, while the average order value was \$1,087.63 . Additionally, the year 2024 was derived with and the month "Jul" was formatted. Formula =SUM(K2:K2001) Formula =AVERAGE(J2:J2001) Formula =YEAR(B2) Formula =TEXT(B2,"MMM")

Outcome	
	1. Missing Quantity values were replaced with a default value of 1. 2. Missing Discount values were replaced with 0 and formatted as percentages. 3. Missing Total Amount values were replaced with the average value of the column. 4. The dataset was cleaned and prepared for accurate analysis and reporting.

Descriptive Statistics Summary (Simple Format)

1. Central Tendency (Average Values)

- **Mean (Average):** 1,276.21
- **Median (Middle Value):** 1,008.27
- **Mode (Most Frequent Value):** 1,541.95

2. Data Spread (Variability)

- **Standard Deviation:** 950.03
- **Range:** 3,971.66
 - Minimum Value: 21.14
 - Maximum Value: 3,992.80
- **Variance:** 902,563.76

3. Distribution Shape

- **Skewness:** 0.82
- **Kurtosis:** -0.13

4. Reliability of the Mean

- **Standard Error:** 21.25
- **95% Confidence Interval:** 1,234.54 to 1,317.88

Interpretation:

- The mean is higher than the median, which shows that some high values are increasing the average. This indicates a **right-skewed (positively skewed)** distribution
- The data values are spread out over a large range, showing high variability.
- Positive skewness indicates the data has more high-value observations.
- Negative kurtosis indicates a slightly flatter distribution with fewer extreme outliers than a normal distribution.
- The true average value of the population is likely to fall between **1,234.54 and 1,317.88** with 95% confidence.
- The dataset shows **high variation in values** and is **positively skewed**, meaning a few larger values increase the average.
- The data is spread across a wide range, but extreme outliers are not very common.
- Overall, the average value is estimated to be around **1,276.21**.
- **Interactive Dashboarding**
 - A central "Dashboard" sheet was created (as seen in the file) that visualizes key findings, likely including:
 - Total Revenue & Profit KPIs.
 - Sales breakdown by **Year, Region, Payment Type, and Product Category**.
 - PivotTables and Charts for dynamic filtering and exploration.

★ Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

1. Total Revenue: **2.55 million**
2. Total Profit: **1.27 million**
3. Highest Revenue Category: **Sports**
4. Highest Profit Category: **Sports**
5. Top Payment Method: **PayPal**
6. Sales increased significantly from **2023 → 2024 → 2025**

➤ **Descriptive analysis:**

- ✳ Sports category generated the highest revenue.
- ✳ Sports category generated the highest profit.
- ✳ PayPal is the most used payment method.
- ✳ Revenue exceeded 2.5 million.

➤ **Diagnostic Analysis:**

- ✳ Sports products have strong demand and higher purchase frequency.
- ✳ Higher discounts may increase sales volume.
- ✳ Loyal customers contribute a significant portion of revenue.

➤ **Predictive Analysis:**

- ✳ Sales grew strongly from 2023 to 2025, indicating continued growth potential.

➤ **Prescriptive Analysis:**

- ✳ Increase inventory for Sports products.
- ✳ Focus marketing on top-selling products.
- ✳ Promote low-performing categories such as Beauty.
- ✳ Expand loyalty programs to encourage repeat purchases.
- ✳ Monitor high-demand products to prevent stockouts.

➤ **Dashboard screenshot**



Conclusion:

- ★ The e-commerce business demonstrates strong growth, with total revenue exceeding 2.5 million and total profit exceeding 1.27 million.
- ★ The Sports category is the leading contributor to both sales and profit.
- ★ Sales have shown a significant upward trend from 2023 to 2025, indicating positive business growth.
- ★ PayPal is the preferred payment method among customers.
- ★ To sustain growth, the business should focus on high-performing categories, optimize inventory management, strengthen customer loyalty programs, and implement targeted promotions for lower-performing categories.